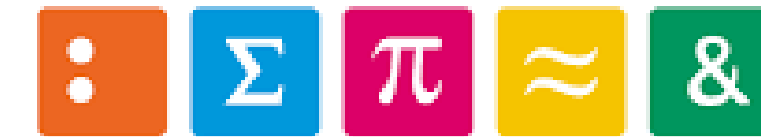




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Goal and Scope

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Goal

This study performs a cradle to grave attributional Life Cycle Assessment (LCA) of a state-of-the-art Proton Exchange Membrane Fuel Cell (PEMFC) stack with a 100 kW electric power output. The study shall conduct a hotspot analysis on the PEMFC stack, which is modeled using a modular structure developed by the experts at HES-SO. The goal of this study is to identify the primary contributors to the potential environmental impact of the overall life cycle at different tiers (as described in the modular structure), thereby establishing a baseline for comparison with future PEMFC stacks.

Riemer in 2023 performed an LCA on a 100kW PEMFC stack and published the Life Cycle Impact Assessment (LCIA) results for the midpoint category as suggest by the ILCD handbook. The Life Cycle Inventory (LCI) in this study shall be compiled based on Riemer's article and its sources, coupled with the assumptions of stack design based on James et al. 2018 (as mentioned in Riemer's article) molded into the modular structure provides an accurate LCI with a structure coherent to that of other hydrogen technologies. This model shall serve as a baseline for the ECOPEM project, which aims to compare the environmental performance of a non-Nafion based membrane PEMFC stack against a Nafion-based one (baseline).





Goal

This study is meant to communicate the main impact contributors of the PEMFC stack to the technical experts of the ECOPEM project. This information will guide them in identifying where to focus their innovation efforts to create a more environmentally friendly design.



Scope

For this cradle to grave Life Cycle Assessment (LCA), the functional unit is defined as 1 kWh of net electricity delivered from a 100 kW Proton Exchange Membrane Fuel Cell (PEMFC) stack operating within Europe during 2023.

The reference flows for this functional unit shall include:

- 1.67×10^{-6} of a 100kW PEMFC stack (36 secs used out of its 6000h lifetime)
- Operation consumables: 0.0547 kg of hydrogen (5.47kg H₂/hr at 2.5 bar) and 3.10kg of air (310.06 kg air/hr at 2.5 bar).
- End of Life of the 100kW PEMFC stack components.

The stages considered in the study and the process tree can be found in the modular structure with two different terminologies in the deliverables folder on google drive.

- Terminologies coherent with Reimer 2023.
- Terminologies coherent with the modular structure ones, used throughout HES-SO for all hydrogen technologies to maintain consistency.



Scope

The modular structure breaks down the process tree into system tiers that further branch into stack tiers. The system tiers, labeled Tier 1 and Tier 2, encompass the entire system, including manufacturing, operational consumables, End of Life (EoL) processes and Balance of Plant (BoP). The stack tiers, labeled Tiers 1 through 5, detail the manufacturing of the main components of the PEMFC stack, which further subdivides into the production of individual sub-components.

The study's Life Cycle Inventory (LCI) will employ the cutoff approach (recycled content approach) for modeling. The primary database will be Ecoinvent-3.11-cutoff, supplemented by a material database developed by HES-SO experts to address gaps, such as for carbon fiber and other specialized materials not found in the Ecoinvent database. Since the PEMFC technology lacks an established recycling concept and mandatory regulations as of 2023, the study notes that the future implementation of closed-loop recycling could significantly reduce environmental impacts.



Scope

The Life Cycle Impact Assessment (LCIA) will be performed using Impact World+ v.2.1 (footprint version). The open-source software Activity Browser (AB) version 2.11.2, which is built on the Brightway2 framework, will be used to conduct the LCA. A sensitivity analysis will then be conducted using Environmental Footprint (EF v.3.1) to verify that the key impact contributors remain consistent. The PEMFC stack's design is based on the "James et al. 2018" reference, as cited in Riemer's article. Detailed material lists, assumptions, and calculations are accessible in the "Deliverables" folder on Google Drive.

This preliminary Life Cycle Assessment (LCA) deliberately cuts-off the Balance of Plant (BoP) components due to a current lack of data. This is a significant limitation, as the BoP is a substantial contributor to the total environmental impact. Its inclusion could fundamentally alter the assessment by accounting the electricity source used to power the BoP and the associated parasitic loads that diminish the net power output. As an output of the system, water is classified as a waste, not a co-product, because its economic value is negligible compared to the functional unit. The low-grade heat generated by this technology is also not valorized but is instead ignored, in alignment with the BoP exclusion and its associated thermal management systems. Future research will integrate these components to provide a more comprehensive assessment.



**Thank
You**